STATATHON 2025

- Problem Statement ID: PS-5
- Problem Statement Title: AI-Powered Semantic Job Role Mapping for NCO-2015
- PS Category- Software/Hardware: Software
- Team ID:8203
- Team Name (Registered on portal): Rising Phoenix

Rising Phoenix

EUREKA - THE SMART AI SEARCH ENGINE

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Unlike traditional keyword-based systems, our solution understands semantic meaning, handles spelling variations, recognizes synonyms, and supports natural and cross language(eg : hinglish,tanglish,etc.) occupation queries across multiple Indian languages, while a graph-layered validation ensures precise and highly accurate outputs.

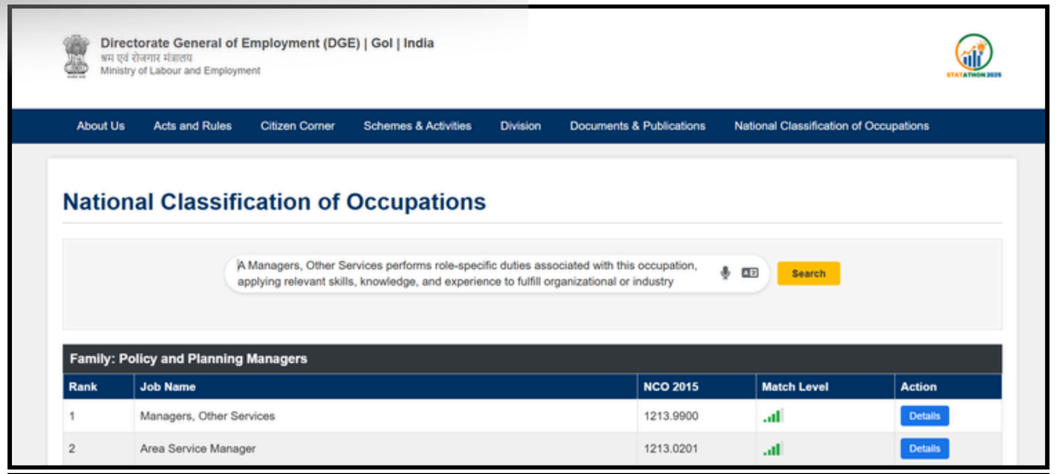
Proposed Solution:

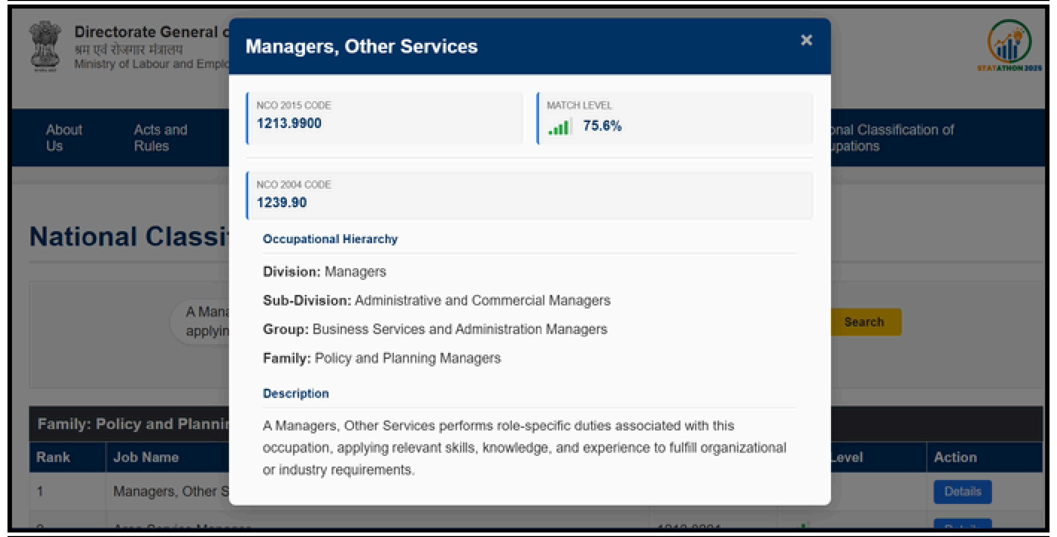
An intelligent AI-enabled system that **understands free-text / voice occupation descriptions** (irrespective of language, grammar, or phrasing) and accurately maps them to NCO-2015 codes using:

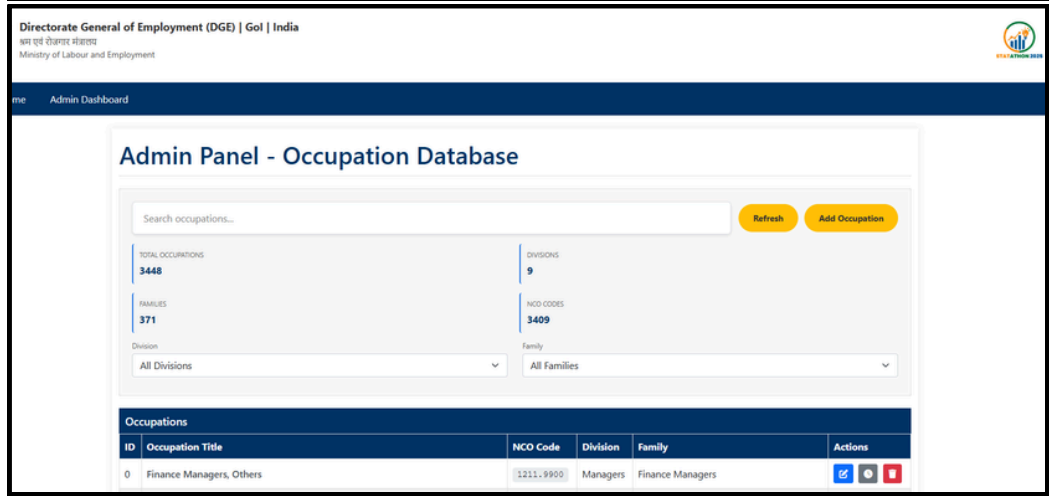
- Automated data processing pipeline to structure and normalize NCO description embeddings
- **SBERT-based** semantic embeddings to capture contextual meaning beyond exact keywords
- **FAISS-powered** vector search for fast and scalable retrieval across all 3,447 occupations
- **Intelligent fallback mechanism** that requests additional user clarification for low-confidence or ambiguous inputs
- **Knowledge-guided Graph Network layer** to enforce hierarchy, domain consistency, and improve overall classification accuracy

Problem Solved:

- Reduces misclassification by **combining semantic intelligence with domain-aware validation**
- Allows users to **input occupations naturally**, without requiring taxonomy knowledge or proper language grammar, and enables queries in **regional languages**
- **Panel for admins** to update and modify the occupation database
- **Minimizes manual intervention** and training effort for enumerators and field officers
- **Handles ambiguity** safely by avoiding forced classification and intelligently prompting for clarification
- Maintains data integrity by **preventing cross-domain errors** through hierarchical and sector-aware validation







Innovation

Prompt Intelligence & Guidance System (P.I.G.S.):

- Detects vague or overly broad user prompts
- Provides context-aware suggestions to refine intent

Clarification Question Generator (C.Q.G.):

- Uses adaptive questions to infer job intent
- Narrows occupations based on user responses

Uniqueness:

- First-of-its-kind **multilingual, graph-guided semantic search** for accurate NCO-2015 occupation mapping
- **Hybrid intelligence pipeline:** semantic embeddings with hierarchical occupation graph validation
- **High-accuracy results** with explainable confidence scores and fallback clarification
- **Continuously improves** through user-selected corrections and feedback loops
- **API as service** for integration with other government applications

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TECHNICAL APPROACH

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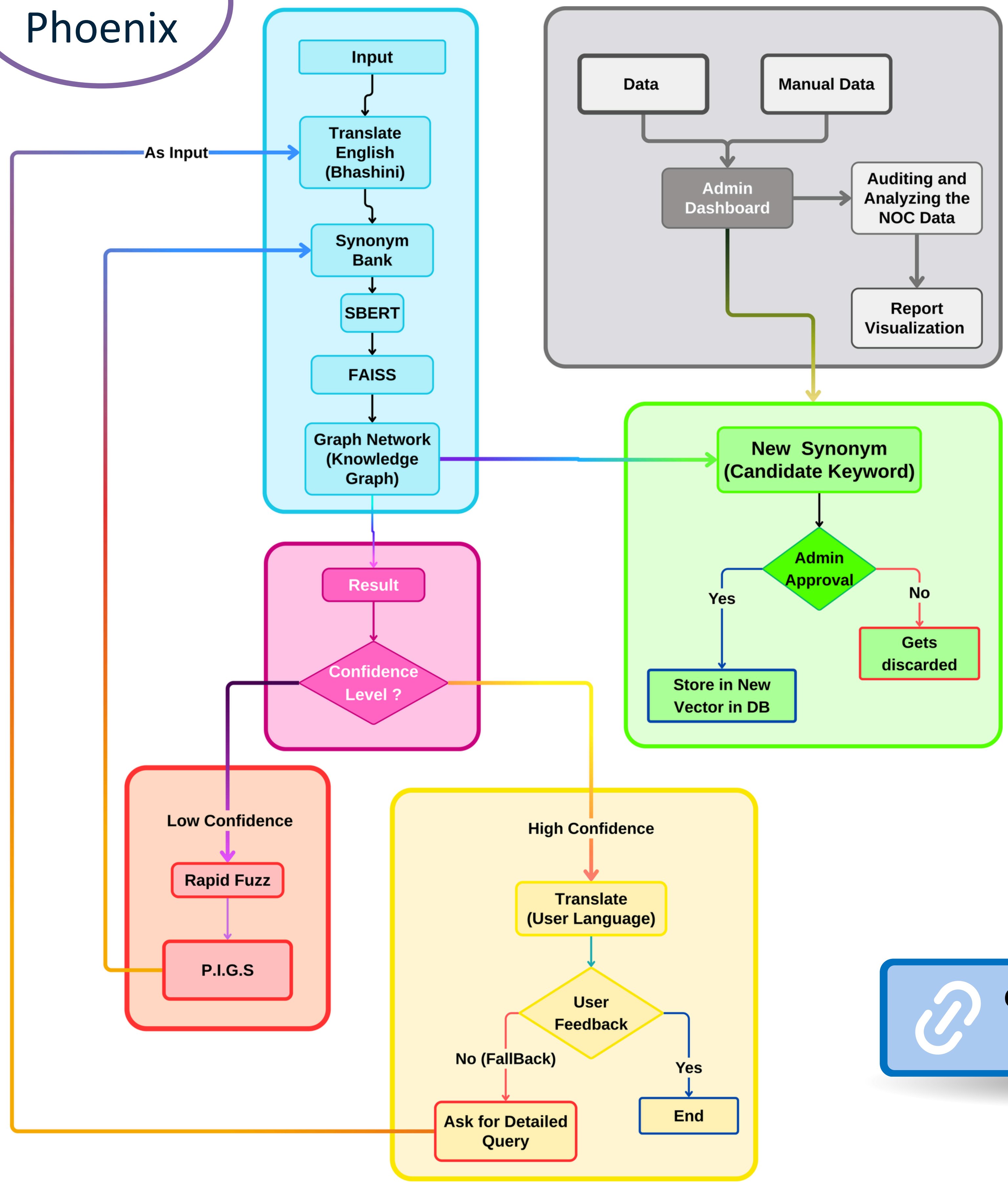

Data Layer

Source: NCO-2015 → structured JSON/CSV.

Database: Stores occupation codes, descriptions, hierarchy.

Embedding Layer

Models: SBERT → semantic embeddings.

Normalization: Ensures consistent floating-point vectors.

Application Layer

Input: User query via **text / voice**.

Language Support: English, Hindi, regional.

Translation: Bhashini, mymemory for unsupported languages.

Output: Result + confidence score.

Eureka - The smart Ai Search Engine

Search Layer

FAISS → fast vector similarity search & ranking.

Graph Network → Forms Graph layer Network for Accuracy and precision

Synonym Bank → maps domain synonyms.

Spelling Correction (Rapid Fizz) → Levenshtein distance / phonetic matching.

GitHub: Explore the Repository

Drive: ProtoType Video

Intelligent Fallback Layer

- **Low-confidence** results trigger query reformulation or clarification prompts
- **P.I.G.S** : **Detects vague** or overly broad user prompts
- Approved new synonyms stored after admin approval, enabling **continuous improvement**

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FEASIBILITY AND VIABILITY

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Feasibility

- **Multilingual by Design –** Bhashini + mymemory → Covers 25+ Indian languages & dialects.
- **Scalable & Efficient –** **FAISS handles millions of survey entries** in near real time and all the included technologies are open source.
- **Lightweight Custom RAG Infra –** Can run on government cloud (NIC) or low-cost AWS instances.

Viability

- **Direct Government Impact –** **Speeds up MoSPI surveys**, reduces manual errors, ensures consistency.
- **Low-Cost Deployment –** Minimal infra, no licensing fees → high ROI.
- **Self-Learning Engine –** Every correction improves accuracy → better over time.
- **Data Security and High Adoption Potential –** Deployable on Govt. secure cloud (MeitY/NIC), ensuring compliance and Works for Census, Labour surveys, NSSO & even state-level surveys.

Potential Challenges:

- Incomplete or noisy data affecting accuracy.
- Ambiguity in job descriptions.

Dependencies:

- NCO 2015 Dataset.
- Bhashini API
- Mymemory

Existing Solution	Proposed Solution
Keyword match only	**Understands semantic and user natural input meaning** by AI intelligent system
Fails with typos/spelling	Handles spelling with native and shortcut words and synonyms
Only English mostly	supports multilingual and **project result in native language**
Slow & less scalable	fast with **FAISS** and **Graph Network** (more Accruate)
User struggles if word unknown	**User Friendly** → text + voice + live translation + natural job description

IMPACT AND BENEFITS

Impact with Benefits of EUREKA

Social Impact

- **Bridges language barriers** → inclusivity for all regions of India
- Helps job seekers understand roles in their native language
- Supports government initiatives for employment standardization

Environmental Impact

- **Digital-first approach** → reduces paper-based job classification
- **Optimized systems** → lower resource consumption

Sustainability:

- The system supports seamless integration of updated NCO datasets with minimal architectural changes. This ensures long-term usability and accuracy as occupational classifications evolve.

Economic Impact

- Faster and more accurate job-role classification → boosts hiring efficiency
- Reduces mismatch between skills and jobs → **increases productivity**
- Cost-effective solution (**AI automation reduces manual work**)

Benefits for India

- Accessible for both **literate, illiterate, semi-literate people.**
- Easy for taking the surveys by **integrating both voice and text system for mulitlingual data**

Innovation & Long-term Benefits

- **AI + multilingual NLP** ensures adaptability for future expansions
- Scalable to new job taxonomies and global standards
- Empowers both employers & employees with transparency and clarity

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RESEARCH AND REFERENCES

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Referred Documents

IndicBERT (AI4Bharat)

- Multilingual ALBERT model trained on 12 Indian languages.
- https://indicnlp.ai4bharat.org/pages/indic-bert

FAISS by Meta AI

- Industry-standard vector similarity search library used for scalable semantic retrieval.
- https://github.com/facebookresearch/faiss

IndoWordNet

- Lexical database covering synonyms & semantic relations across 18 Indian languages.
- https://en.wikipedia.org/wiki/IndoWordNet

Bhashini API:

- Government of India’s language technology APIs for translation and speech services.
- https://bhashini.gov.in/api-documentation

Research Papers

- Taxonomy-guided Semantic Indexing for Academic Paper Search : https://aclanthology.org/2024.emnlp-main.407/
- Performant and Predicate-Agnostic Search Over Vector Embeddings and Structured Data: https://arxiv.org/abs/2403.04871
- General Filtered Search across Vector and Structured Data: https://arxiv.org/abs/2510.27141
- A Native Relational Database for Hybrid Queries on Structured and Unstructured Data: https://arxiv.org/abs/2501.05006

Industrial People Contacted

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Admin Panel - Occupation Database

Search occupations...

Refresh Add Occupation

TOTAL OCCUPATIONS
3448

DIVISIONS
9

FAMILIES
371

NCO CODES
3409

Division
All Divisions

Family
All Families

Occupations

ID	Occupation Title	NCO Code	Division	Family	Actions
0	Finance Managers, Others	1211.9900	Managers	Finance Managers	